

# BTS\_Transformation — dbt Transformation Layer

How `stg_flights` is transformed into the `FLIGHT_CORE` Fact Constellation (`dim_date`, `dim_airport`, `dim_airline`, `fact_flight`, `fact_flight_operation`, `fact_flight_delay`)

This document explains the dbt transformation models that turn the cleansed `stg_flights` staging model (already a union of `RAW_FLIGHTS_2024/2025/2026`) into the mart-layer Fact Constellation schema in `BTS_AIRLINE_DB.FLIGHT_CORE`.

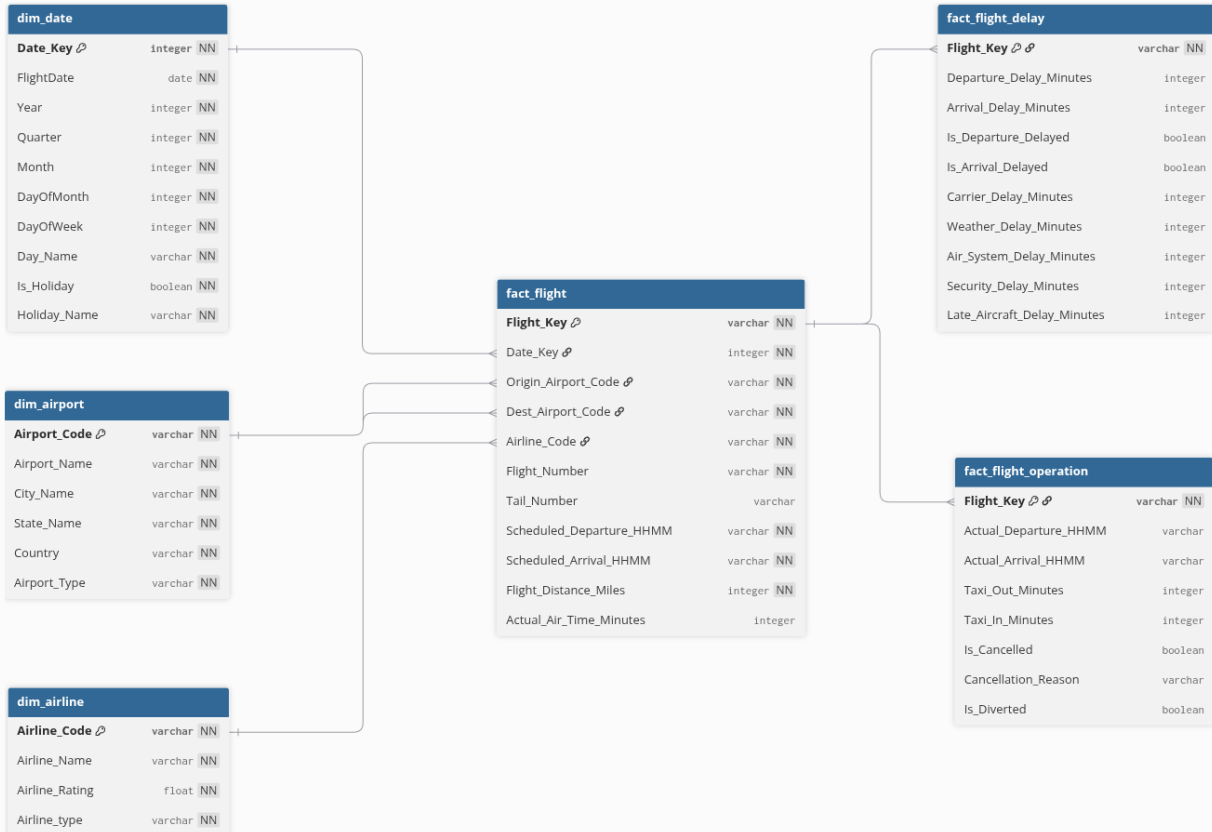
## Overall Approach

The schema is a Fact Constellation (Galaxy Schema): three fact tables share the same grain (one row per flight) and are linked to three conformed dimensions. `dim_airport` is a single role-playing dimension used for both the origin and the destination side of a flight.

- `dim_date`, `dim_airport`, `dim_airline` — built as regular (full-refresh) models
- `fact_flight`, `fact_flight_operation`, `fact_flight_delay` — incremental models, merge strategy, `unique_key = Flight_Key`

# Schema Diagram

The FLIGHT\_CORE Fact Constellation: three conformed dimensions on the left (dim\_date, dim\_airport, dim\_airline), the core fact\_flight in the center, and the two satellite facts (fact\_flight\_delay, fact\_flight\_operation) on the right, all linked through Flight\_Key.



# 1. Dimensions

## dim\_date

Built from the distinct calendar attributes already present in `stg_flights` (Year, Quarter, Month, DayOfMonth, DayOfWeek).

- `Date_Key` is generated as an INTEGER in YYYYMMDD format — it is derived directly from the date rather than an arbitrary surrogate key, which keeps it human-readable and sortable.
- `Day_Name` comes from Snowflake's `DAYNAME()` function.
- `Is_Holiday` combines two conditions: weekends (`DayOfWeek IN (6,7)`) OR a match in the `US_federal_holidays` seed, joined on the holiday date.
- `Holiday_Name` resolves to the actual holiday name, 'Weekend', or 'No Holiday' depending on which condition matched.

## dim\_airport (role-playing dimension)

Airports appear twice in every flight record (origin and destination), but the project models a single conformed dimension rather than two separate ones.

- Two CTEs (`origin_airports`, `dest_airports`) pull the origin and destination columns from `stg_flights`, renamed to shared column names (`Airport_Code`, `City_Name`, `State_Name`).
- A `UNION` (not `UNION ALL`) merges them into one deduplicated list of every airport that appears anywhere in the flight data, on either side.
- This combined list is then `LEFT JOINed` to `stg_airport_info` (sourced from the OurAirports JSON) to enrich it with `Airport_Name`, `Country`, and `Airport_Type`.
- `COALESCE` provides defaults ('Unknown Airport', 'United States', 'Unknown Airport Type') for airports with no match in the reference data, so the join never produces NULLs in those columns.

At query time, `fact_flight` joins to this single dimension twice — once aliased as `origin` and once as `dest` — which is the standard pattern for a role-playing dimension.

## dim\_airline

- Distinct `Airline_Code` values are pulled from `stg_flights`.
- `LEFT JOINed` to `stg_airline_info` (sourced from an airline metadata JSON) to attach `Airline_Name`, `Airline_Type`, and `Airline_Rating` (from Skytrax).

## 2. Fact Tables

All three facts share the same grain and the same surrogate key, `Flight_Key`, generated upstream in `stg_flights`. Splitting one wide fact into three narrower ones keeps the core flight identity separate from operational status and from delay metrics, which simplifies testing and keeps each model focused on one concern.

### **fact\_flight — the core fact**

- The only one of the three facts that joins directly to the dimensions: `dim_date` (on `FlightDate`), `dim_airport` twice (origin and destination), and `dim_airline`.
- Carries the foreign keys (`Date_Key`, `Origin_Airport_Code`, `Dest_Airport_Code`, `Airline_Code`) plus core flight attributes: `Flight_Number`, `Tail_Number`, scheduled times, distance, and actual air time.
- Incremental filter compares `FlightDate` against the maximum `Date_Key` already loaded in the table, converting the stored `YYYYMMDD` integer back to a `DATE` with `TO_DATE(TO_VARCHAR(MAX(Date_Key)), 'YYYYMMDD')`.

### **fact\_flight\_operation — operational status**

- No dimension joins at all — it only relates to `fact_flight` through `Flight_Key`.
- Holds actual departure/arrival times, taxi times, and flight status: `Is_Cancelled`, `Cancellation_Reason`, `Is_Diverted`.
- An `INNER JOIN` to `fact_flight` enforces that every row here corresponds to a flight that already exists in the core fact — referential integrity by construction.
- Incremental logic differs from `fact_flight`: instead of filtering by date, it selects every `Flight_Key` already present in `fact_flight` that is NOT YET present in this table (`WHERE Flight_Key NOT IN (SELECT Flight_Key FROM {{ this }})`).

### **fact\_flight\_delay — delay metrics**

- Same pattern as `fact_flight_operation`: no dimension joins, `INNER JOIN` to `fact_flight`, and the same `NOT IN` incremental filter on `Flight_Key`.
- Holds `Departure_Delay_Minutes`, `Arrival_Delay_Minutes`, the delay flags (`Is_Departure_Delayed`, `Is_Arrival_Delayed`), and the delay breakdown by cause (`Carrier`, `Weather`, `Air System`, `Security`, `Late Aircraft`).

### **Why split into three facts instead of one wide fact?**

- Separation of concerns — identity/dimensions, operational status, and delay metrics are modeled and tested independently.
- Performance — a query that only needs delay analysis does not have to scan every column of the full flight record.
- Same grain throughout — all three tables share one row per flight and one `Flight_Key`, so this is a vertical split of one logical fact across three physical tables, not three facts at different grains.

### 3. Data Tables (Column-Level)

Column-level structure of each dimension and fact table, as defined in FLIGHT\_CORE. PK = primary key, FK = foreign key.

#### dim\_date

| Column              | Type    | Key | Notes                                    |
|---------------------|---------|-----|------------------------------------------|
| <b>Date_Key</b>     | integer | PK  | Generated as YYYYMMDD                    |
| <b>FlightDate</b>   | date    | -   | Calendar date                            |
| <b>Year</b>         | integer | -   |                                          |
| <b>Quarter</b>      | integer | -   | 1-4                                      |
| <b>Month</b>        | integer | -   |                                          |
| <b>DayOfMonth</b>   | integer | -   |                                          |
| <b>DayOfWeek</b>    | integer | -   | 1=Mon ... 7=Sun                          |
| <b>Day_Name</b>     | varchar | -   | From Snowflake DAYNAME()                 |
| <b>Is_Holiday</b>   | boolean | -   | Weekend OR US federal holiday            |
| <b>Holiday_Name</b> | varchar | -   | Holiday name, 'Weekend', or 'No Holiday' |

#### dim\_airport (role-playing dimension)

| Column              | Type    | Key | Notes                                      |
|---------------------|---------|-----|--------------------------------------------|
| <b>Airport_Code</b> | varchar | PK  | Deduplicated across origin and destination |
| <b>Airport_Name</b> | varchar | -   | Defaults to 'Unknown Airport'              |
| <b>City_Name</b>    | varchar | -   |                                            |
| <b>State_Name</b>   | varchar | -   |                                            |
| <b>Country</b>      | varchar | -   | Defaults to 'United States'                |
| <b>Airport_Type</b> | varchar | -   | Defaults to 'Unknown Airport Type'         |

#### dim\_airline

| Column                | Type    | Key | Notes               |
|-----------------------|---------|-----|---------------------|
| <b>Airline_Code</b>   | varchar | PK  | Unique carrier code |
| <b>Airline_Name</b>   | varchar | -   |                     |
| <b>Airline_Rating</b> | float   | -   | From Skytrax        |
| <b>Airline_type</b>   | varchar | -   |                     |

### fact\_flight (core fact)

| Column                   | Type    | Key | Notes                                            |
|--------------------------|---------|-----|--------------------------------------------------|
| Flight_Key               | varchar | PK  | Surrogate key, generated upstream in stg_flights |
| Date_Key                 | integer | FK  | -> dim_date                                      |
| Origin_Airport_Code      | varchar | FK  | -> dim_airport                                   |
| Dest_Airport_Code        | varchar | FK  | -> dim_airport                                   |
| Airline_Code             | varchar | FK  | -> dim_airline                                   |
| Flight_Number            | varchar | -   |                                                  |
| Tail_Number              | varchar | -   | Nullable                                         |
| Scheduled_Departure_HHMM | varchar | -   | Local time, HHMM                                 |
| Scheduled_Arrival_HHMM   | varchar | -   | Local time, HHMM                                 |
| Flight_Distance_Miles    | integer | -   |                                                  |
| Actual_Air_Time_Minutes  | integer | -   | Nullable                                         |

### fact\_flight\_delay (satellite fact)

| Column                      | Type    | Key     | Notes          |
|-----------------------------|---------|---------|----------------|
| Flight_Key                  | varchar | PK / FK | -> fact_flight |
| Departure_Delay_Minutes     | integer | -       |                |
| Arrival_Delay_Minutes       | integer | -       |                |
| Is_Departure_Delayed        | boolean | -       |                |
| Is_Arrival_Delayed          | boolean | -       |                |
| Carrier_Delay_Minutes       | integer | -       |                |
| Weather_Delay_Minutes       | integer | -       |                |
| Air_System_Delay_Minutes    | integer | -       |                |
| Security_Delay_Minutes      | integer | -       |                |
| Late_Aircraft_Delay_Minutes | integer | -       |                |

### fact\_flight\_operation (satellite fact)

| Column     | Type    | Key     | Notes          |
|------------|---------|---------|----------------|
| Flight_Key | varchar | PK / FK | -> fact_flight |

| Column                       | Type    | Key | Notes                                        |
|------------------------------|---------|-----|----------------------------------------------|
| <b>Actual_Departure_HHMM</b> | varchar | -   | Local time, HHMM                             |
| <b>Actual_Arrival_HHMM</b>   | varchar | -   | Local time, HHMM                             |
| <b>Taxi_Out_Minutes</b>      | integer | -   |                                              |
| <b>Taxi_In_Minutes</b>       | integer | -   |                                              |
| <b>Is_Cancelled</b>          | boolean | -   |                                              |
| <b>Cancellation_Reason</b>   | varchar | -   | Decoded inline, no separate dim_cancellation |
| <b>Is_Diverted</b>           | boolean | -   |                                              |

## 4. Model Summary

| Model                 | Type                     | Materialization     | Joins To                              |
|-----------------------|--------------------------|---------------------|---------------------------------------|
| dim_date              | Dimension                | Table / View        | US_federal_holidays (seed)            |
| dim_airport           | Dimension (role-playing) | Table / View        | stg_airport_info                      |
| dim_airline           | Dimension                | Table / View        | stg_airline_info                      |
| fact_flight           | Fact (core)              | Incremental (merge) | dim_date, dim_airport x2, dim_airline |
| fact_flight_operation | Fact (satellite)         | Incremental (merge) | fact_flight (INNER JOIN)              |
| fact_flight_delay     | Fact (satellite)         | Incremental (merge) | fact_flight (INNER JOIN)              |

### Incremental Strategy Recap

- fact\_flight: date-driven high-water-mark against its own Date\_Key.
- fact\_flight\_operation / fact\_flight\_delay: key-driven — load any Flight\_Key that fact\_flight already has and this table does not, then merge on Flight\_Key.
- All three use on\_schema\_change='fail', so an unexpected upstream column change stops the run instead of silently altering the table.